

Backpaper - Probability II (2024-25)

Time: 3 hours.

Attempt all questions. The total marks is 50.

1. The joint density function of X and Y is given by

$$f(x, y) = xe^{-x(y+1)}, \quad x > 0, y > 0.$$

- (a) Find the conditional density of X , given $Y = y$. [4 marks]
(b) Find the joint density function of $Z = XY$ and $W = \frac{X}{Y}$. [4 marks]
2. Let X_1, X_2, X_3 be independent $U(0, 1)$ random variables. Let $Z = \max(X_1, X_2, X_3)$.

- (a) Find the CDF of Z . [3 marks]
(b) Find $E(Z)$. [4 marks]

3. State the Strong Law of Large Numbers. [3 marks]

4. (a) State the Central Limit Theorem. [3 marks]

- (b) Let $X_1, X_2, \dots, X_{100}$ be i.i.d. random variables with $\mathbf{P}(X_1 = 1) = 0.6$, $\mathbf{P}(X_1 = -1) = 0.4$. Give an approximate value of the probability $P(X_1 + X_2 + \dots + X_{100} < 10)$ in terms of $\Phi(x)$ (the CDF of the $N(0, 1)$ distribution) using the Central Limit Theorem. [4 marks]

5. The random variables X and Y have joint density function

$$f(x, y) = 12xy(1 - x) \quad 0 < x < 1, 0 < y < 1,$$

and equal to 0 otherwise.

- (a) Are X and Y independent? [1 mark]
(b) Find $E(X)$. [3 marks]
(c) Find $P(X + Y \leq 1)$. [3 marks]
6. Consider a simple random walk on $\mathbf{Z}$ with probability of going to the right equal to 0.6 and probability of going to the left equal to 0.4. Suppose the walk starts at 50. What is the probability that the walk hits 0 before 100? [7 marks]
7. In a sequence of Bernoulli trials with success (S) probability p and failure (F) probability $q = 1 - p$, let a_n be the probability that the combination SF occurs for the first time at trials $n - 1$ and n . Let T be the random variable such that $P(T = n) = a_n$.
- (a) Find the generating function of T . [7 marks] HINT: condition on first trial.
(b) Find the mean of T . [4 marks]